Set 21: Energy Consumption Analysis

DSA – Data Handling and Visualization | Lab Experiment

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Aim

To analyse an energy consumption dataset in R using a histogram and density plot, a transparent bubble scatter plot and a sector-wise bar chart, and to build a filtered interactive dashboard in Tableau.

Software / Tools Used

R 4.3.3 with ggplot2 (charts), gridExtra (tables) and patchwork (dashboard layout); Tableau Desktop / Tableau Public (interactive dashboard).

Dataset

Energy_Consumption_Data.csv

Sector	Region	Month	Temperature (°C)	Units_Consumed (kWh)	Cost (₹)	Renewable_Usage (%)	Peak_Hours
Residential	North	Jan	15	320	2100	22	4
Commercial	South	Jan	24	540	3600	18	6
Industrial	West	Feb	20	880	5900	12	8
Residential	East	Feb	18	350	2300	25	5
Commercial	North	Mar	28	610	4100	20	7
Industrial	South	Mar	30	920	6200	15	9

Program – Dataset Creation (common to all R tasks)

```
library(ggplot2)
library(gridExtra)
library(patchwork)
options(scipen = 999)
# Plain white table theme with black borders (no colours)
tbl_theme <- ttheme_minimal(
  core    = list(fg_params = list(col = "black", fontsize = 11),
                 bg_params = list(fill = "white", col = "black")),
  colhead = list(fg_params = list(col = "black", fontface = "bold", fontsize = 11.5),
                 bg_params = list(fill = "white", col = "black")))
# Create Energy_Consumption_Data.csv and import it into R
raw <- "Sector,Region,Month,Temperature,Units_Consumed,Cost,Renewable_Usage,Peak_Hours
Residential,North,Jan,15,320,2100,22,4
Commercial,South,Jan,24,540,3600,18,6
Industrial,West,Feb,20,880,5900,12,8
Residential,East,Feb,18,350,2300,25,5
Commercial,North,Mar,28,610,4100,20,7
Industrial,South,Mar,30,920,6200,15,9"
writeLines(raw, "Energy_Consumption_Data.csv")
energy <- read.csv("Energy_Consumption_Data.csv", stringsAsFactors = FALSE)
energy$Month  <- factor(energy$Month, levels = c("Jan", "Feb", "Mar"))
energy$Sector <- factor(energy$Sector, levels = c("Residential", "Commercial", "Industrial"))
pal <- c(Residential = "#009E73", Commercial = "#E69F00", Industrial = "#0072B2")
```

Q1. Import, Histogram and Density Plot of Units_Consumed

> *Import the dataset in R. Plot a histogram and density plot for Units_Consumed. Compare consumption patterns.*

R Program:

```
str(energy)
cat("\nSummary of Units_Consumed:\n"); print(summary(energy$Units_Consumed))
cat("Mean:", round(mean(energy$Units_Consumed), 1), "| Median:",
median(energy$Units_Consumed),
    "| SD:", round(sd(energy$Units_Consumed), 1), "\n")
cat("\nAverage Units_Consumed by Sector:\n")
print(aggregate(Units_Consumed ~ Sector, data = energy, FUN = mean))
p_hist <- ggplot(energy, aes(x = Units_Consumed)) +
  geom_histogram(aes(y = after_stat(density)), binwidth = 200, boundary = 200,
                 fill = "#0072B2", alpha = 0.6, colour = "white") +
  geom_density(colour = "#D55E00", linewidth = 1.2, adjust = 1) +
  scale_x_continuous(limits = c(0, 1200), breaks = seq(0, 1200, 200)) +
  labs(title = "Histogram and Density of Units Consumed", x = "Units Consumed (kWh)", y =
"Density") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
print(p_hist)
ggsave("s21_q1.png", p_hist, width = 7, height = 4.3, dpi = 150, bg = "white")
p_sec <- ggplot(energy, aes(x = Units_Consumed, y = Sector, colour = Sector)) +
  geom_point(size = 5, alpha = 0.8) +
  ggrepel::geom_text_repel(aes(label = paste(Region, Month)), size = 3.2, colour = "black",
nudge_y = 0.25, seed = 2) +
  scale_colour_manual(values = pal) + scale_x_continuous(limits = c(250, 1000)) +
  labs(title = "Units Consumed by Sector", x = "Units Consumed (kWh)", y = NULL) +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"),
legend.position = "none")
ggsave("s21_q1b.png", p_sec, width = 7, height = 3.6, dpi = 150, bg = "white")
```

Console Output:

```
'data.frame':  6 obs. of  8 variables:
 $ Sector         : Factor w/ 3 levels "Residential",..: 1 2 3 1 2 3
 $ Region         : chr  "North" "South" "West" "East" ...
 $ Month          : Factor w/ 3 levels "Jan","Feb","Mar": 1 1 2 2 3 3
 $ Temperature    : int  15 24 20 18 28 30
 $ Units_Consumed : int  320 540 880 350 610 920
 $ Cost           : int  2100 3600 5900 2300 4100 6200
 $ Renewable_Usage: int  22 18 12 25 20 15
 $ Peak_Hours     : int  4 6 8 5 7 9

Summary of Units_Consumed:
   Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
  320.0   397.5   575.0   603.3   812.5   920.0
Mean: 603.3 | Median: 575 | SD: 255.1

Average Units_Consumed by Sector:
       Sector Units_Consumed
1 Residential            335
2  Commercial            575
3  Industrial            900
```

Output:

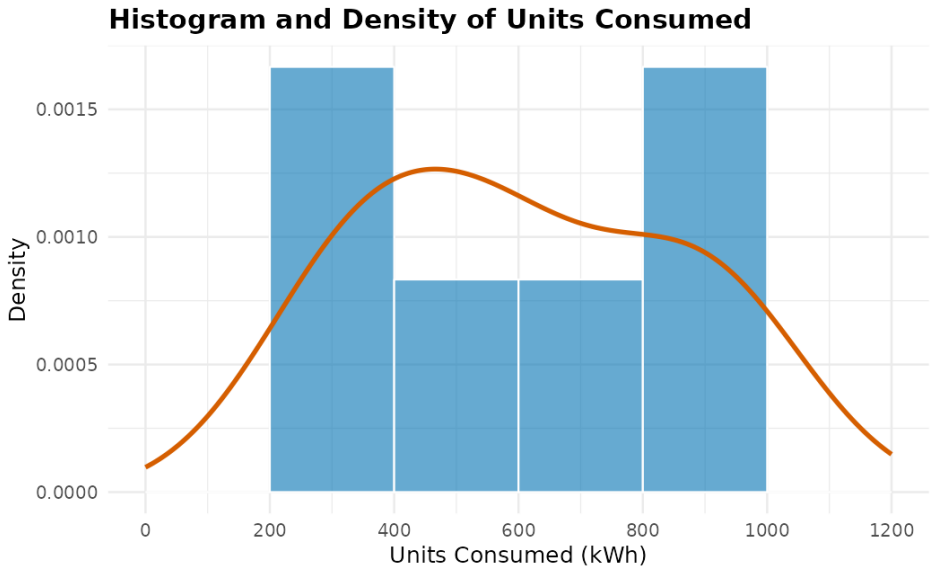


Figure 1: Histogram with density curve of Units_Consumed

Output:

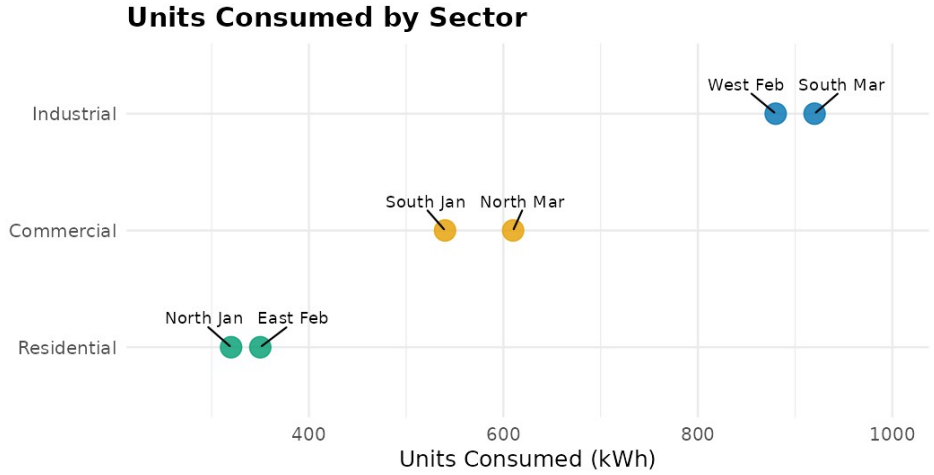


Figure 2: Units consumed by sector

Inference:

Units consumed range from 320 to 920 kWh, with a mean of 603.3 and a median of 575 (standard deviation 255.1). The density curve has **three clusters**: a low group (320, 350), a middle group (540, 610) and a high group (880, 920).

The clusters match the sectors. **Residential** averages 335 kWh, **Commercial** 575 kWh and **Industrial** 900 kWh, so Industrial users consume about 2.7 times as much as Residential users. Consumption is driven mainly by sector, and the sample is small (6 records), so the smooth curve should be read as indicative.

Q2. Scatter Plot – Temperature vs Units_Consumed with Bubble Size

> *Using R programming, create a scatter plot between Temperature and Units_Consumed. Represent Peak_Hours using bubble size. Apply transparency.*

R Program:

```
cat("Correlation Temperature vs Units_Consumed:", round(cor(energy$Temperature,
energy$Units_Consumed), 3), "\n")
cat("Correlation Peak_Hours vs Units_Consumed :", round(cor(energy$Peak_Hours,
energy$Units_Consumed), 3), "\n")
fit <- lm(Units_Consumed ~ Temperature, data = energy)
```

```
cat("Regression: Units =", round(coef(fit)[1], 1), "+", round(coef(fit)[2], 1), "* Temperature
(R-squared =", round(summary(fit)$r.squared, 3), ")\n")
p_sc <- ggplot(energy, aes(x = Temperature, y = Units_Consumed)) +
  geom_point(aes(size = Peak_Hours, colour = Sector), alpha = 0.6) +
  geom_text(aes(label = paste(Region, Month)), vjust = -1.8, size = 3.2) +
  scale_size_area(max_size = 14, breaks = c(4, 6, 8), name = "Peak Hours") +
  scale_colour_manual(values = pal) +
  scale_x_continuous(limits = c(12, 33)) + scale_y_continuous(limits = c(200, 1050)) +
  labs(title = "Temperature vs Units Consumed (bubble size = Peak Hours)", x = "Temperature (\
u00B0C)", y = "Units Consumed (kWh)") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold"))
print(p_sc)
ggsave("s21_q2.png", p_sc, width = 7.5, height = 4.8, dpi = 150, bg = "white")
```

Console Output:

```
Correlation Temperature vs Units_Consumed: 0.655
Correlation Peak_Hours vs Units_Consumed : 0.977
Regression: Units = -38.2 + 28.5 * Temperature (R-squared = 0.429 )
```

Output:

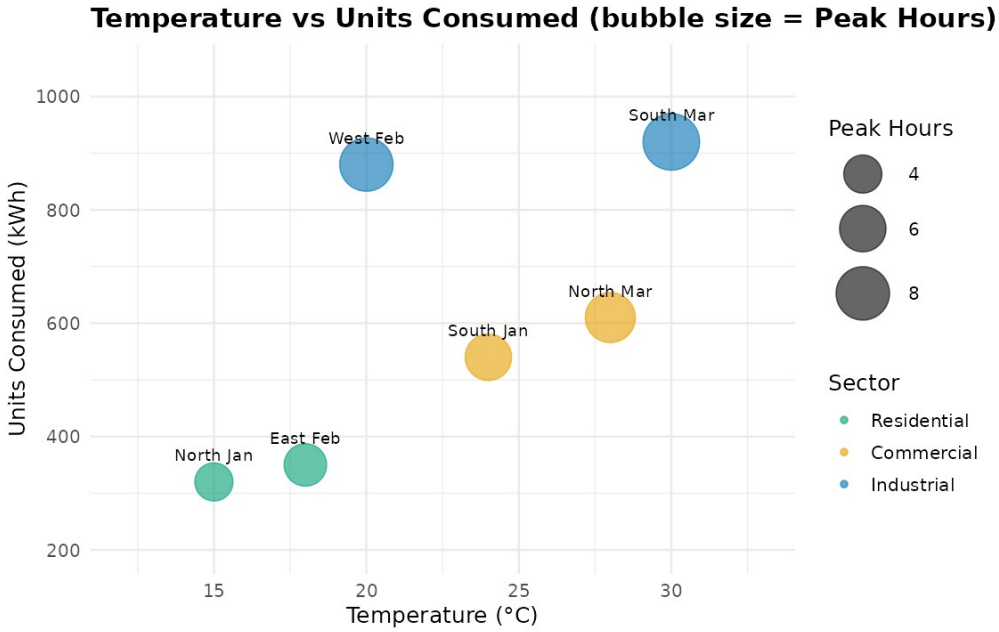


Figure 3: Temperature vs units consumed; bubble size = Peak Hours; alpha = 0.6

Inference:

Temperature and consumption have a moderate positive relationship (r = 0.655, R^2 = 0.43): hotter months and places use more energy. The relationship is far from perfect, because sector matters more than temperature: for example, the 20 °C Industrial record (880 kWh) uses more than the 28 °C Commercial record (610 kWh).

Peak hours track consumption very closely (r = 0.977). The largest bubbles (8–9 peak hours) are the Industrial records at the top right (880 and 920 kWh), and the smallest bubbles (4–5 hours) are the Residential records at the bottom left. Longer peak-demand periods go with much higher energy use.

Q3. Average Renewable_Usage by Sector

> *Using R, calculate average Renewable_Usage for each Sector. Create a bar chart. Interpret renewable energy adoption.*

R Program:

```
avg_ren <- aggregate(Renewable_Usage ~ Sector, data = energy, FUN = mean)
names(avg_ren)[2] <- "Avg_Renewable_Usage"
print(avg_ren, row.names = FALSE)
cat("Overall average:", round(mean(energy$Renewable_Usage), 2), "%\n")
p_ren <- ggplot(avg_ren, aes(x = Sector, y = Avg_Renewable_Usage, fill = Sector)) +
  geom_col(width = 0.55) +
  geom_text(aes(label = paste0(Avg_Renewable_Usage, "%")), vjust = -0.5, size = 4.5) +
  scale_fill_manual(values = pal) + scale_y_continuous(limits = c(0, 30)) +
  labs(title = "Average Renewable Usage by Sector", x = "Sector", y = "Average Renewable Usage
(%)") +
  theme_minimal(base_size = 12) + theme(plot.title = element_text(face = "bold", hjust = 0.5),
legend.position = "none")
print(p_ren)
ggsave("s21_q3.png", p_ren, width = 6.5, height = 4.3, dpi = 150, bg = "white")
```

Console Output:

```
      Sector Avg_Renewable_Usage
 Residential                23.5
  Commercial                19.0
  Industrial                13.5
Overall average: 18.67 %
```

Output:

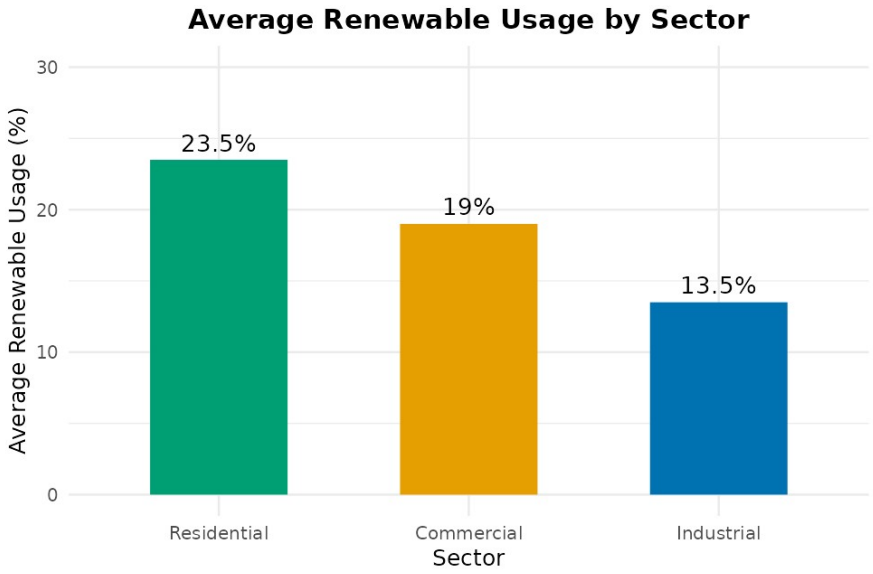


Figure 4: Average renewable usage per sector

Inference:

Average renewable usage is **Residential 23.5%**, **Commercial 19.0%** and **Industrial 13.5%** (overall average 18.67%).

Renewable adoption is **inversely related to consumption**: the sector that uses the least energy (Residential) has the highest renewable share, and the biggest consumer (Industrial) has the lowest. Industrial users therefore offer the largest opportunity to increase renewable adoption, since a small rise in their share would cover many more kWh.

Q4. Tableau – Line Chart, Heatmap, Filters and Interactive Dashboard

> *Import Energy_Consumption_Data.csv into Tableau. Create a line chart for monthly energy usage trends. Create a heatmap (Sector vs Region vs Units). Add filters for Month and Sector. Design an interactive dashboard.*

R Program:

```
monthly <- aggregate(Units_Consumed ~ Month, data = energy, FUN = sum)
print(monthly, row.names = FALSE)
p_trend <- ggplot(monthly, aes(x = Month, y = Units_Consumed, group = 1)) +
  geom_line(colour = "#0072B2", linewidth = 1.2) + geom_point(size = 4, colour = "#0072B2") +
  geom_text(aes(label = Units_Consumed), vjust = -1.2, size = 4) +
  scale_y_continuous(limits = c(700, 1700)) +
  labs(title = "Monthly Energy Usage Trend", x = "Month", y = "Total Units Consumed (kWh)") +
  theme_minimal(base_size = 11) + theme(plot.title = element_text(face = "bold"))
heat <- aggregate(Units_Consumed ~ Sector + Region, data = energy, FUN = sum)
p_heat <- ggplot(heat, aes(x = Region, y = Sector, fill = Units_Consumed)) +
  geom_tile(colour = "white", linewidth = 1) +
  geom_text(aes(label = Units_Consumed, colour = Units_Consumed > 650), size = 4.2) +
  scale_colour_manual(values = c("TRUE" = "white", "FALSE" = "black"), guide = "none") +
  scale_fill_gradient(low = "#DEEBF7", high = "#08306B", name = "Units (kWh)") +
  labs(title = "Heatmap: Sector vs Region (Units Consumed)", x = "Region", y = "Sector") +
  theme_minimal(base_size = 11) + theme(plot.title = element_text(face = "bold"), panel.grid =
element_blank())
dash <- (p_trend | p_heat) +
  plot_annotation(title = "Energy Consumption Dashboard (filters: Month, Sector)",
                  theme = theme(plot.title = element_text(face = "bold", size = 15)))
ggsave("s21_dash.png", dash, width = 12, height = 4.8, dpi = 130, bg = "white")
```

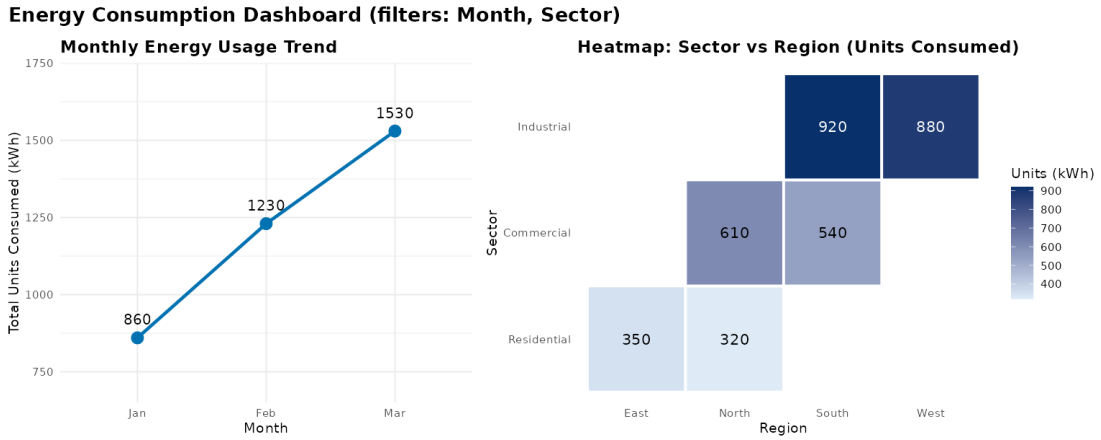
Console Output:

```
 Month Units_Consumed
   Jan            860
   Feb           1230
   Mar           1530
```

Procedure in Tableau:

1. Open Tableau → **Connect → Text file** → **Energy_Consumption_Data.csv**. Set Month as a Dimension (String) and create a custom sort Jan, Feb, Mar (right-click Month → Sort → Manual).
2. **Sheet 1 – "Monthly Trend":** drag Month to Columns and SUM(Units_Consumed) to Rows. Set the Marks type to **Line** with markers and labels. Expected totals: Jan 860, Feb 1,230, Mar 1,530.
3. **Sheet 2 – "Sector-Region Heatmap":** drag Region to Columns, Sector to Rows and SUM(Units_Consumed) to **Color** and Label. Set the Marks type to **Square** and choose a sequential colour palette.
4. **Filters:** drag Month and Sector to the **Filters** shelf → right-click → **Show Filter**. Right-click each → **Apply to Worksheets → All Using This Data Source**.
5. **Dashboard → New Dashboard:** place the line chart on the left and the heatmap on the right, with the Month and Sector filter cards in a vertical container. Add a title.
6. Set both sheets to **Use as Filter**. Clicking a heatmap cell then filters the trend line to that sector and region. Save as **.twbx** or publish to Tableau Public.

Dashboard Layout:



Reference layout of the dashboard (drawn in R). The interactive version is built in Tableau using the procedure above.

Inference:

Total energy use grows every month: 860 kWh in January, 1,230 in February (+43.0%) and 1,530 in March (+24.4%). The heatmap shows that the darkest cells are the Industrial records (South 920, West 880). Filtering by Sector or Month lets users see which combinations drive the increase.

Result

The energy consumption data was analysed in R with a histogram and density plot, a transparent bubble scatter plot and a sector-wise renewable usage bar chart, and an interactive Tableau dashboard was designed. Consumption is driven by sector (Industrial about 900 kWh vs Residential 335 kWh), peak hours track consumption (r = 0.98), and renewable adoption is highest in the lowest-consuming sector.